LESSON 136 FLIGHT (1.5 DUAL, and 0.3 INSTRUMENT)

Lesson Objective

The purpose of this lesson is to review the procedures and maneuvers from stage two in preparation for the stage check. This lesson will be used to evaluate the student's ability to conduct a VFR cross country as Pilot in Command as well as explain weather and airspace.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
  - Review maneuvers to be performed in 136.
  - Plan the cross-country route of the flight to be flown in 136.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Normal and Crosswind Takeoff

- Ensure runway line-up check is performed.
- The student must look at the windsock prior to establishing a crosswind correction.
- Verify proper rudder usage.
- Verify rotation happens smoothly at 55 knots.
- Crosswind correction should be kept up until rotation.
- On the departure leg, the student should track the runway depending on the winds.

Use of Mixture

- Have the student lean for best power.
  - Ensure the correct leaning procedure is performed.
- Have the student lean for best economy.
  - Ensure the correct leaning procedure is performed.

Pilotage and Dead Reckoning

- Departure:
  - Set up radios, GPS, navaids, etc.
  - TIME OFF must be recorded.
  - Radio Communications:
    - Simulate opening the flight plan with FSS.
    - Simulate getting flight following with approach.
- Cruise:
  - Make sure the student is switching tanks and leaning.
  - Evaluate CRM and division of attention.
  - Pilotage & Dead Reckoning:
    - The student looks for checkpoints.
    - Course must be intercepted.
    - ATAs recorded.
    - Record actual Ground Speed.
    - Calculate revised ETA.
    - Verify location.
    - Complete the nav log.

VOR Tracking

- For the purpose of this lesson, the student must be able to:
  - Set the VOR frequency.
  - Tune and ID the VOR.
  - Set the HSI to VLOC mode.
  - Figure out the current radial.
  - Intercept and track inbound/outbound instructions using an intercept angle of 30-45 degrees in the correct direction.

Diversion Procedures

- Simulate a scenario where a diversion is necessary (unforecasted weather, system malfunction, instrument malfunction, engine complications, lost comms, etc.)
- The student must confirm the present position on the sectional chart.
- Choose an alternate shown on the sectional or use the 'Nearest' page on the GPS.
- The student must divert immediately toward the alternate using shortcuts/rule of thumb calculations if it is an emergency. If time permits, the student may circle until all of the information is obtained.

- The diversion log must be filled out.
- The chart supplement must be used to obtain the required information for the alternate airport.
- Stress aircraft control.
- The student must show situational awareness.
- The flight plan should be closed.

Basic Attitude Instrument Flight/Unusual Attitudes

- The student must be able to perform the following:
    - Straight and level flight.
    - Standard rate turns and medium level turns.
    - Constant airspeed climbs and descents.
    - Recoveries from unusual attitudes (ensure that the student recovers using the correct procedure depending on nose low or nose high).

Steep Turns

- Student does, student says:
    - Verify correct procedure.
    - Ensure that the student is looking both outside and inside.
    - Verify the roll out happens at ½ the bank angle before the desired heading.

Non-Towered Airport Operations

- To increase efficiency, divert the student to a non-towered airport.
- The student must:
    - Obtain the weather.
    - Figure out the entry procedures (45-degree entry to the downwind).
    - Make radio calls within 10NM from the airport and in each leg of the pattern.
    - Conduct the descent and arrival checklist.
    - Continuously scan for traffic.

Short-Field Takeoff and Maximum Performance Climb

- Student does, student says:
    - Verify correct procedure.
    - Ensure that the student holds the brakes until full static RPM.
    - Verify that the climb out is conducted at 63 knots.
    - Verify that the student says “obstacle cleared” if there is an obstacle or simulated obstacle.
    - Verify that the student says positive rate of climb before retracting the flaps.

Short-Field Approach and Landing

- Student does, student says:
    - Verify correct procedure.
    - Emphasize the importance of not forcing the airplane on the ground. If the point cannot

    be made, perform a go-around.
    - Verify simulated maximum breaking.

Soft-Field Takeoff and Climb

- Student does, student says:
    - Verify correct procedure.
    - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
    - Verify that the student says “obstacle cleared” if there is an obstacle or simulated obstacle.
    - Verify that the student says positive rate of climb before retracting the flaps.

Soft-Field Approach and Landing

- Student does, student says:
    - Verify correct procedure.
    - Verify minimal braking.

ATC Communications

- The student must be making all radio communication with ATC.

Checklist Usage

- Ensure that the student uses all pertinent checklists throughout the flight.

Emergency Operations

- This should be coupled with the diversion.
- Give the student an emergency situation resulting in the need to divert.
- Ensure that the checklists are used.

Normal and Crosswind Landing

- Verify correct approach speed.
- The student must conduct a landing where the main touch down before the nose wheel.
- At least one of the main wheels should be on the centerline.
- Ensure that the student is maintaining the proper wind correction.
- The student must be able to judge when to round out and flare.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.

- Preview the next lesson → 137a.
- Assign homework:
    - Study for the stage 2 check and plan one of the TOLD card problems and cross-country route for the stage 2 check (FIT HUB).

Completion Standards

The student shall demonstrate the ability to plan and safely conduct cross-country flights. During the flight the student shall demonstrate, without instructor guidance:

1. Proper navigation procedures,
2. During cross-country: altitude ±200 feet, heading ±15°, position within ±3 NM and ETA within ±5 minutes.
3. Directional control at all times during takeoffs and landings.
4. A normal/crosswind landing and touch down at or within 400 ft beyond a specified point.
5. A short-field landing and touch down at or within 300 ft beyond a specified point.
6. A soft-field landing and touch down softly with no drift.
7. Steep turns while maintaining altitude ±150 ft, airspeed ±15 kts, bank angle ±10° and roll out on the entry heading ±15°.